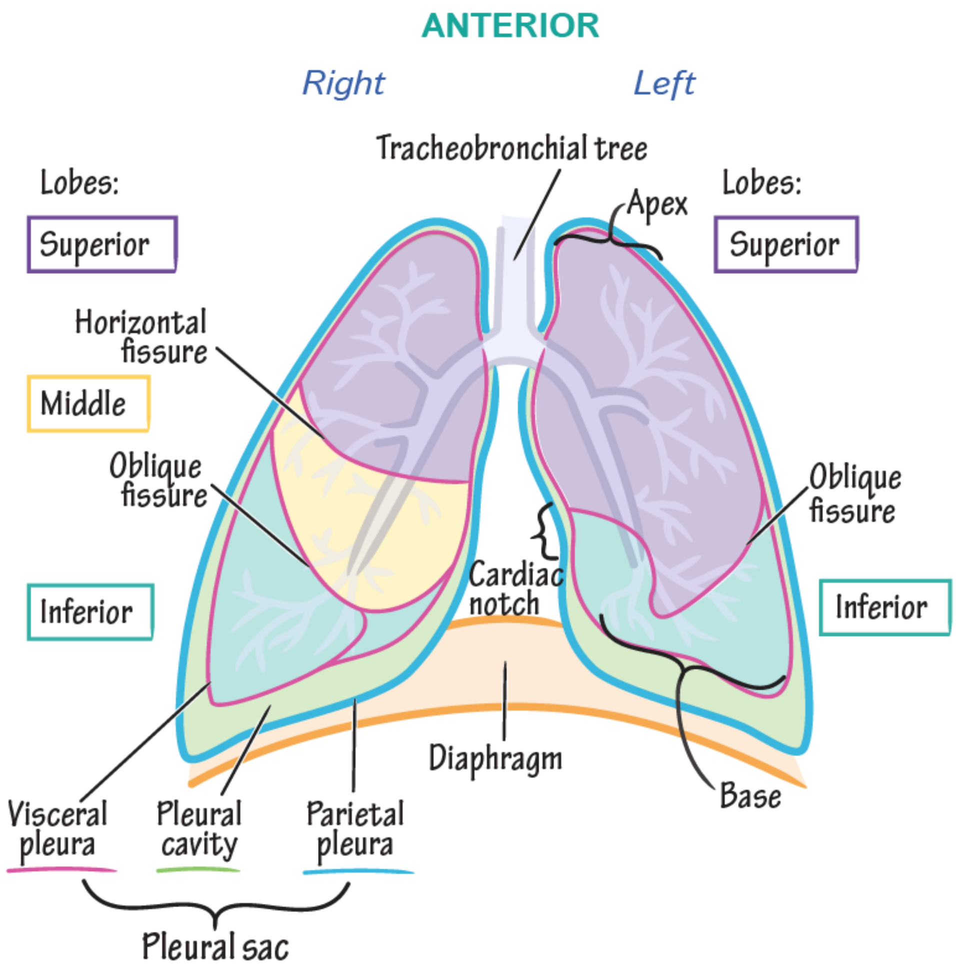
Pleural Disorders

Professor Dr Aram Baram MD, MRCSEd, FACS

Anatomy and physiology



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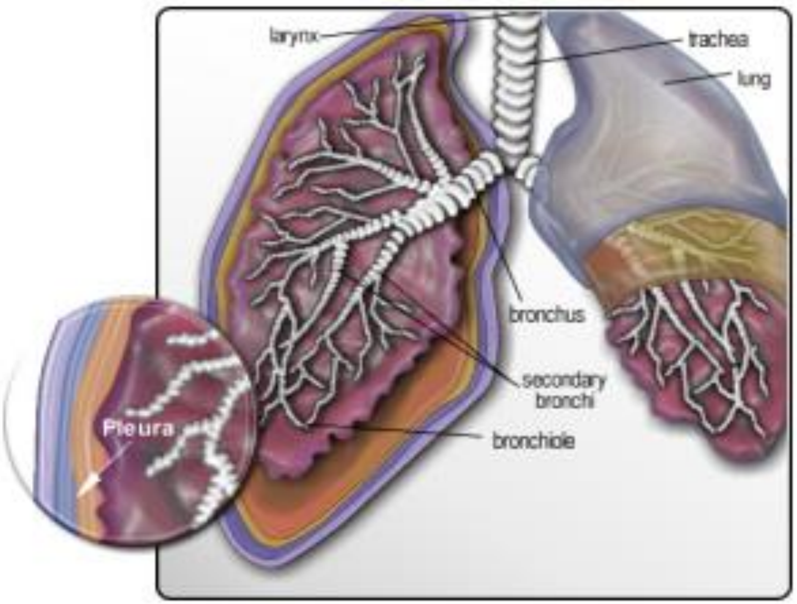
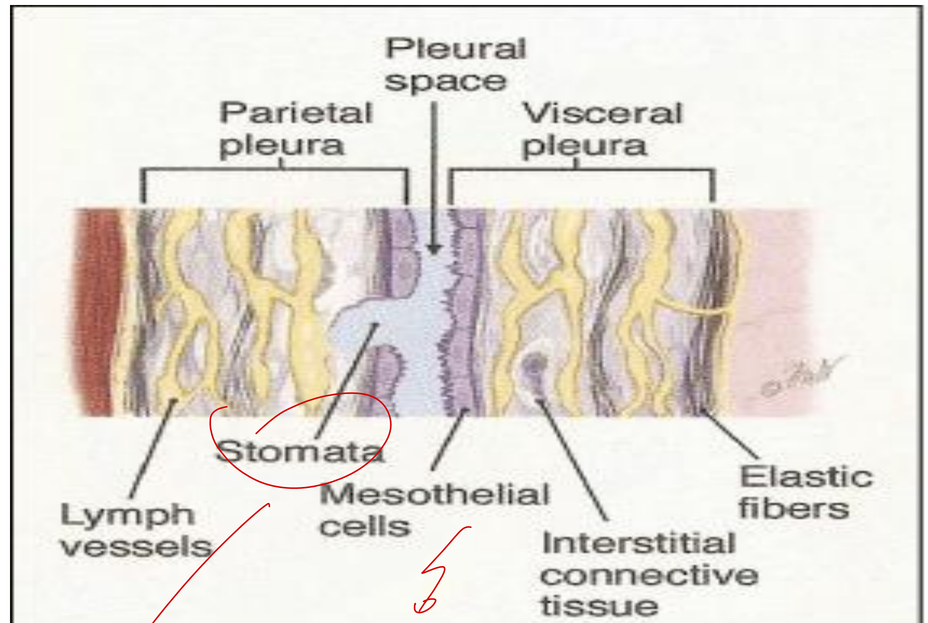
Pleura

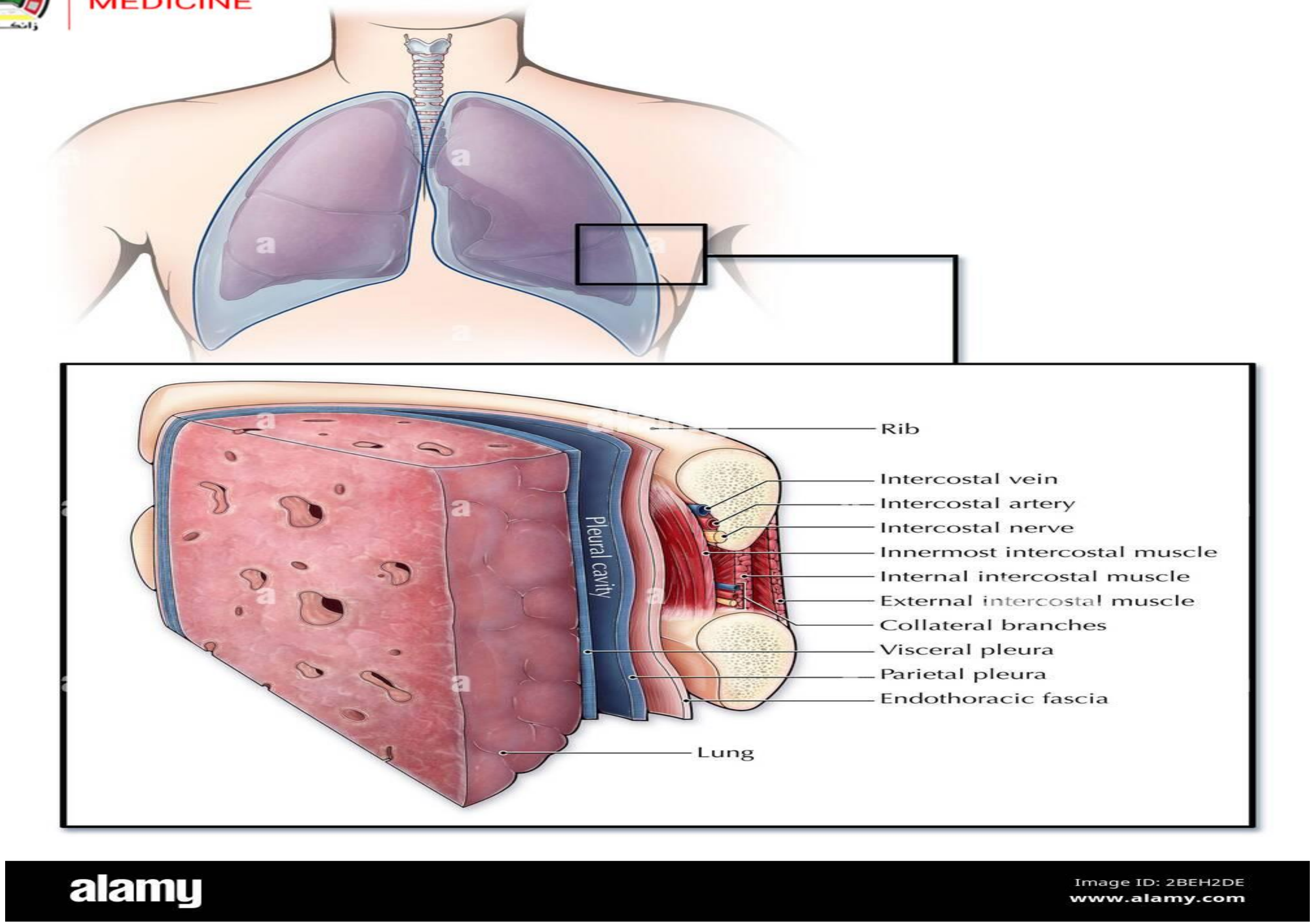
simple layer of mesothelium

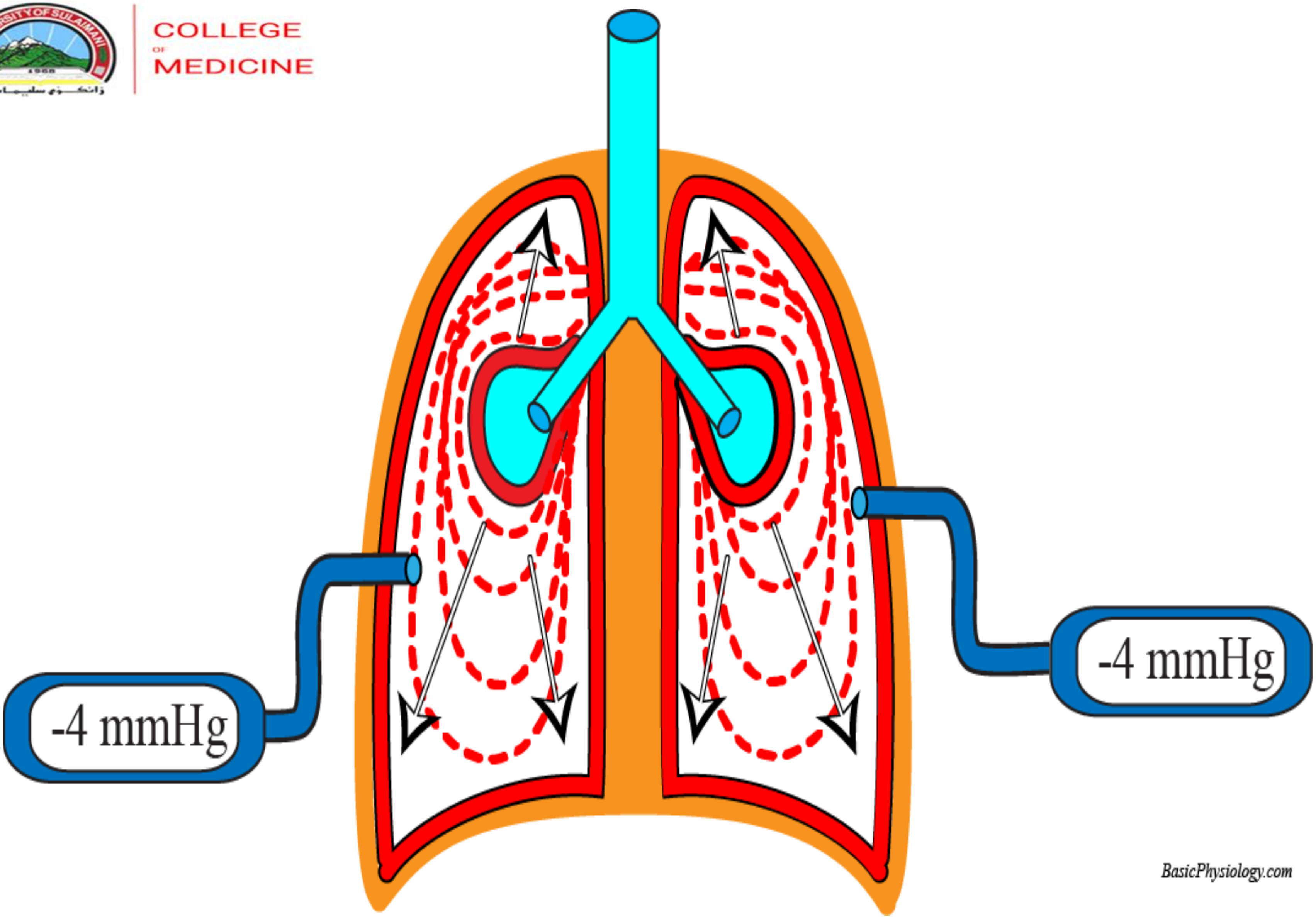
- Parietal
- Visceral





serous fluid


Rib
Intercostal vein
Intercostal artery
Intercostal nerve
Innermost intercostal muscle
Internal intercostal muscle
External intercostal muscle
Collateral branches
Visceral pleura
Parietal pleura
Endothoracic fascia
Pleural cavity
Lung
alamy
Image ID: 2BEH2DE
www.alamy.com


-4 mmHg
-4 mmHg
BasicPhysiology.com

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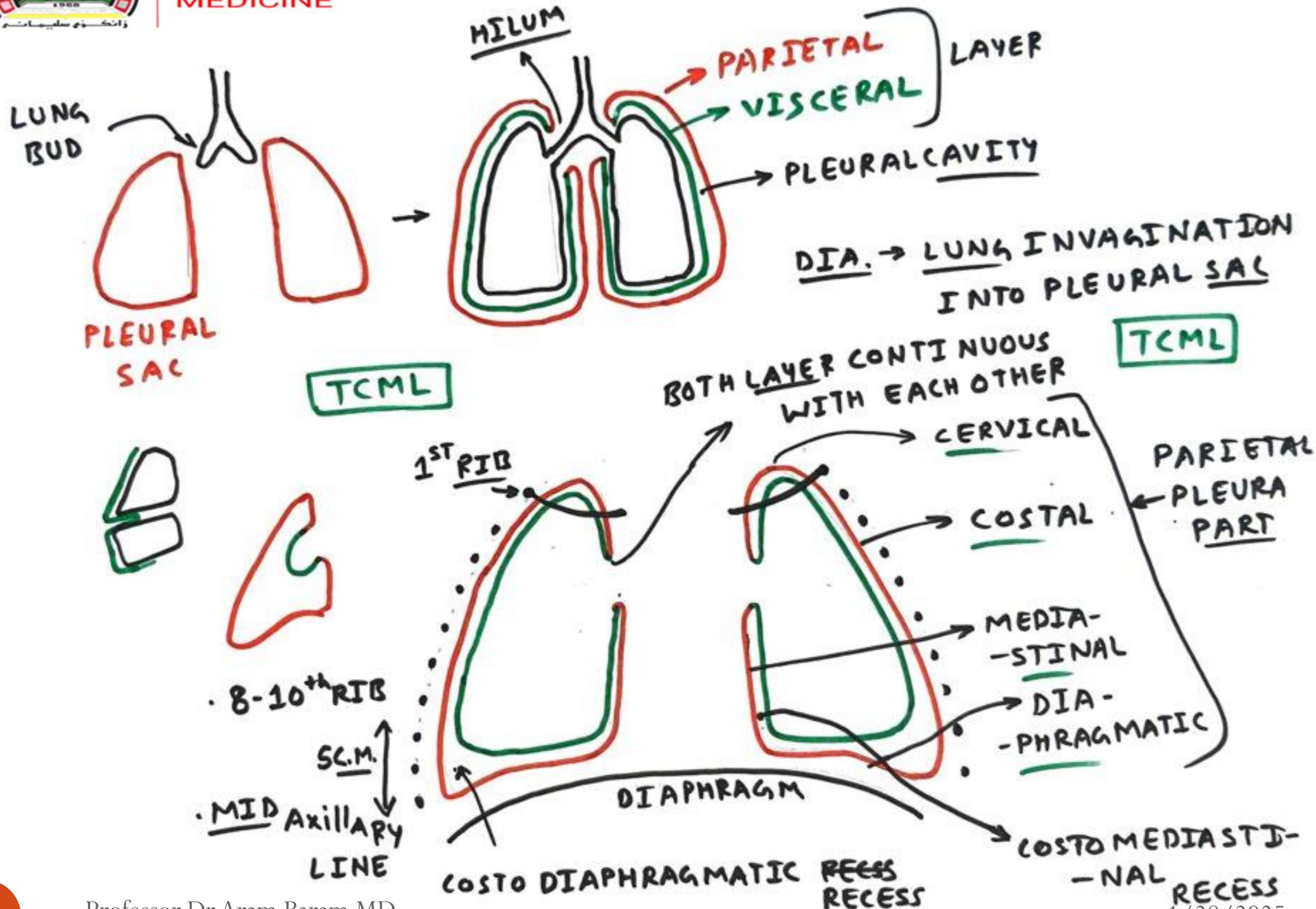
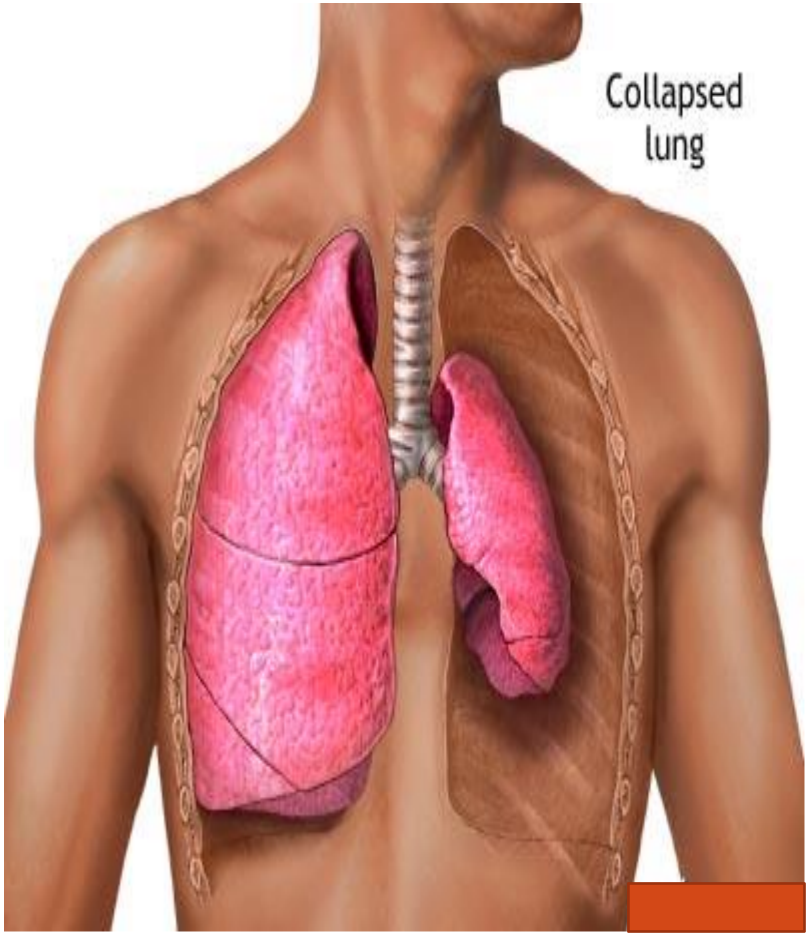


Pneumothorax

- Is an accumulation of air in the pleural space leading to lung collapse.
- Pneumothorax is classified based on their etiology or clinical presentation.
- Normally no air found in the pleural cavity



Classification of Pneumothorax

I. Spontaneous

1) Primary: usually due to sub-pleural bleb rupture. Male predominance of 3:1 is reported, with tall, thin males between the ages of 15 and 30 most commonly affected.

Cigarette smoking increases the risk of primary pneumothorax by a factor of 20.

Primary pneumothorax is more common than secondary pneumothorax.

2) Secondary : Bullous disease like Emphysema, Cystic fibrosis, Connective tissue diseases especially Marfan's syndrome Interstitial lung diseases especially eosinophilic granuloma , Pneumonia with lung abscess, Catamenial pneumothorax, Metastatic cancers especially sarcomas , Lung cancer, Esophageal perforation, Neonatal due to congenital cystic lung disases

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Classification of Pneumothorax

II. Acquired

- **Iatrogenic :** Central line placement, Pacemaker insertion, Transthoracic needle biops, Transbronchial needle biopsy, Thoracocentesis ,After laparoscopic surgery, Barotrauma,
- **Traumatic:** Blunt trauma: Motor vehicle accidents

  Penetrating trauma: Gunshot wounds,

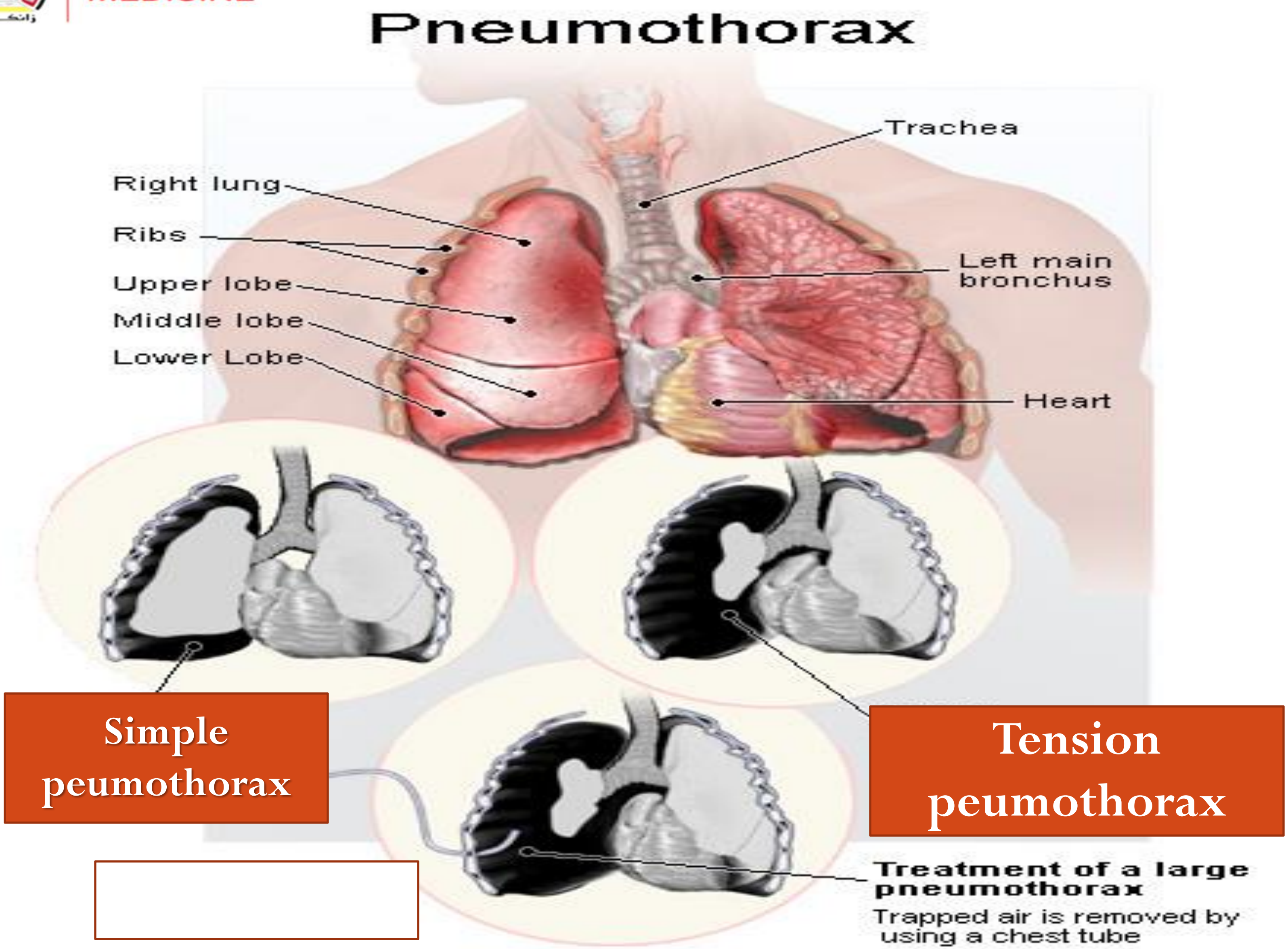
  Stab wounds

TYPES : according to the clinical presentation

1) Simple pneumothorax less than 20%

2) Tension penumothorax total lung collaps, shifting of the mediastinum, haemodynamic, cetral cynosis instablity

3) Open pneumothorax Whenever chest wall wounds present


Pneumothorax
Trachea
Right lung
Ribs
Upper lobe
Middle lobe
Lower Lobe
Left main bronchus
Heart
Simple peumothorax
Tension peumothorax
Treatment of a large pneumothorax
Trapped air is removed by using a chest tube

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Clinical features

- Most cases of spontaneous pneumothorax present with sudden-onset ipsilateral pleuritic chest pain with some shortness of breath and occur when the patient has been at rest.
- Physical examination could normal, particularly for a small pneumothorax (<15% of the hemithorax).
- Patients presenting with larger pneumothoraces may have dyspnea and tachycardia.
- Findings on respiratory examination may include diminished or absent breath sounds, hyperresonant percussion note, decreased movement of the chest wall, and diminished fremitus on the affected side.

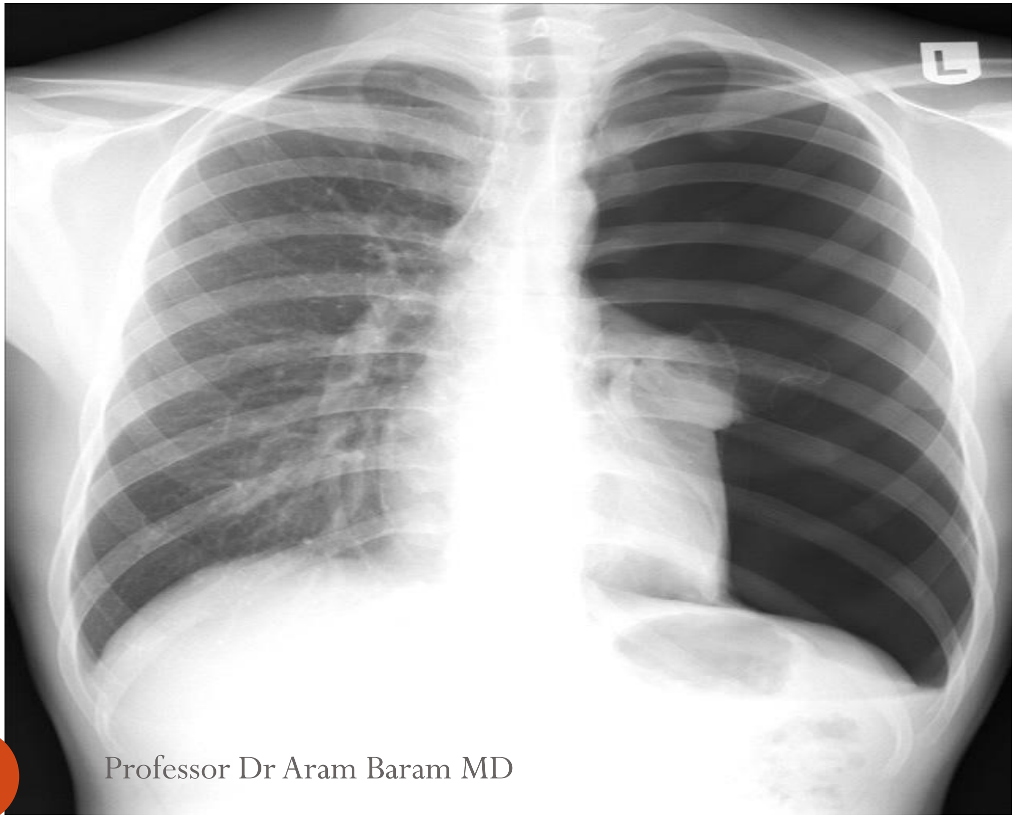
Sign and symptoms of Tension Pneumothorax

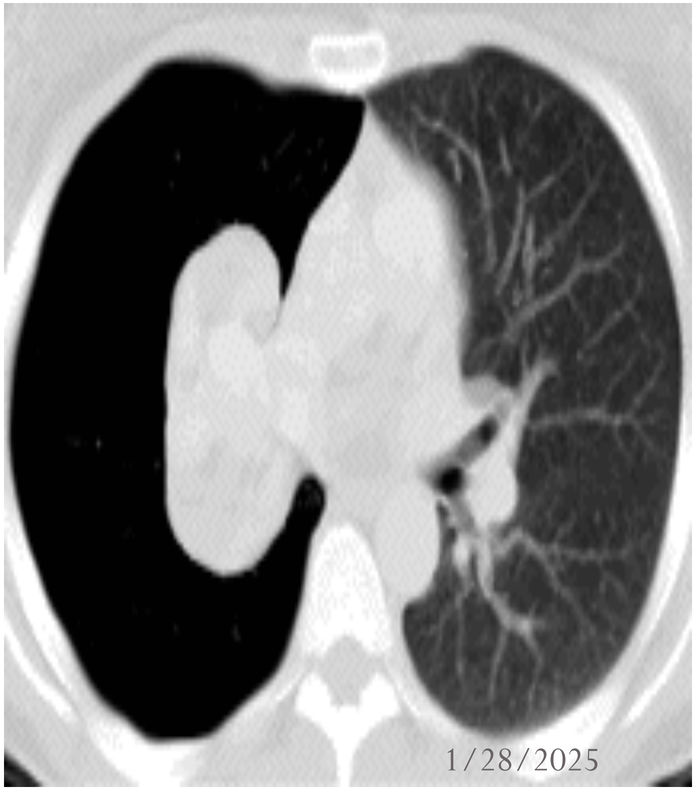
- Restlessness
- Severe Dyspnea
- Absent Breath sounds on affected side
- Tachypnea
- Tachycardia
- Poor Color
- Accessory Muscle Use
- JVD
- Narrowing Pulse Pressures
- Hypotension
- Tracheal Deviation

(late if seen at all)

Investigations :

- 1) A chest radiograph: presence of a thin visceral pleural line displaced from the chest wall on the upright posterior-anterior chest radiograph.
- 2) Occasionally a computed tomography (CT) scan of the chest is helpful in planning surgical intervention.





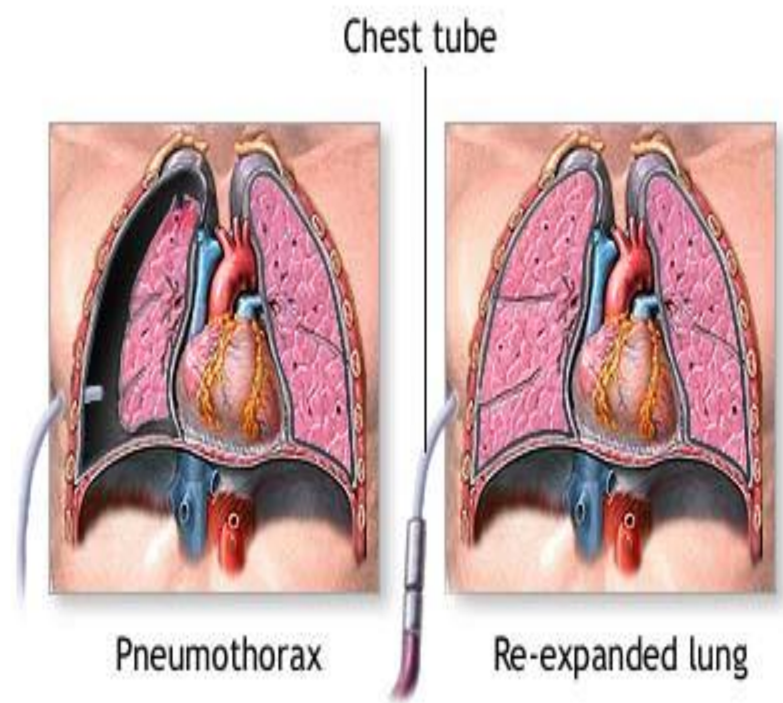
INITIAL TREATMENT :

• *Clinically Stable Patient*

- Observation is reserved for asymptomatic patients who present with a pneumothorax of less than 20% of the pleural cavity.
- In such a case, the patient should be observed in the emergency department for 4–6 hours and discharged home if a repeat chest radiograph excludes progression of the pneumothorax.
- On discharge, the patient should be provided with careful instructions for follow-up within 12–48 hours, depending on circumstances. A repeat chest radiograph is obtained at the follow-up visit to document resolution of the pneumothorax.
- If the patient lives a distance away from emergency services or is considered unreliable, it is safest to admit this patient to the hospital for observation and follow-up chest radiographs.

Tube Thoracostomy

- Larger pneumothoraces (>20%) usually are treated with prompt reexpansion of the lung by tube thoracostomy.
- A tube larger than 26 Fr rarely is required.
- Normally the tube is placed under generous local anesthesia (1% or 2% lidocaine) in the fourth or fifth intercostal space in the anterior or midaxillary line and is directed toward the apex of the affected hemithorax.
- The tube is left in place until the air leak resolves and the lung is fully expanded.



Indications for Surgery :

(1) recurrent pneumothorax: 2^{rd} attack ipsilateraly or 1^{st} attack controlateraly

(2) persistent air leak or incomplete reexpansion of the lung,

(3) massive air leak with incomplete reexpansion of the lung,

(4) occupational hazard or possible lack of access to medical care,

(5) history of tension pneumothorax or prior pneumonectomy, and

(6) hemopneumothorax with persistent bleeding.

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OPERATION

- 1) Video-assisted thoracic surgery (VATS) is the usual way of treating
  2) Other operative approaches to treat spontaneous pneumothorax included thoracotomy (anterior, lateral, or transaxillary).
- Regardless of the technique selected, the bullae should be resected and obtain pleural symphysis either by parietal pleurectomy or by pleural abrasion is key to the success of the procedure.
- Postoperatively the chest tubes are kept to suction for at least 24 hours

SECONDARY PNEUMOTHORACES

- Patients with secondary pneumothoraces generally have significant comorbid diseases and are debilitated from a respiratory standpoint and require that treatment be individualized.
- Treatment should include chemical or surgical pleurodesis in combination with complete lung reexpansion to seal air leaks.
- Tube thoracostomy alone is associated with a high recurrence rate.

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Empyema Thoracis

Empyema Thoracis

DEFINITION

Empyema, from the Greek, is defined simply as "pus in the pleural cavity."The precursor of empyema is bacterial pneumonia and subsequent parapneumonic effusion.

Causes:

1. **Pulmonary infections :** Complicated and unresolved pneumonia, Bronchiectasis, Tuberculosis, Lung abscess, ruptured hydatid cyst
2. **Aspiration of pleural effusion** for any cause may end emypema if infected
3. **Trauma:** Blunt and penetrating chest trauma, thoracic surgery, oesophgeal surgery ,
4. Extra pulmonary causes: Subphrenic abscess

Clinical features:

- There are symptoms of pus at any site like swinging pyrexia with generalize malaise, prulent productive coughy, chest pain.
- Parapneumonic effusions occur in 20–60% of patients hospitalized for bacterial pneumonia; 5–10% of these parapneumonic effusions progress to empyema.
- The mortality rate of empyema is significant: 25% in the elderly and debilitated.

Parapneumonic effusions are exudative and progress through three stages to an empyema:

- **1. *First* or *exudative stage*,** characterized by relatively low LDH, normal glucose, and normal pH
- **2. *Second* or *fibropurulent stage*** evolves with invasion of pleural fluid by bacteria, increased fibrin deposition, cellular debris, and white blood cells with the ultimate formation of limiting fibrin membranes producing loculations
- **3. *Third* or *organization stage*** occurs as fibroblasts grow into the exudative fibrin sheet coating the visceral and parietal pleura with an inelastic membrane or pleural "peel" encasing the lung and rendering it functionless.

Investigations:

1.**Haematological** Leucocytosis, elevated ESR

2.**Standard posterior-anterior** and lateral chest radiographe shows pleural effusion, hydropeumothorax, fibrothorax in late stages

Ultrasonography ***is widely*** available in most institutions and frequently employed after the initial chest X-ray.

Ultrasound is rapid, portable, and less expensive than a chest CT scan which is some time is necessary to identify the loculation

Pleural fluid first must be determined as exudative or transudative,

which is accomplished by examination of pleural fluid obtained by thoracentesis criteria based on levels of lactate dehydrogenase (LDH) and protein concentrations : an exudate as having any one of the following characteristics:

(i) pleural fluid protein divided by serum protein concentration greater than 0.5,

(ii) pleural fluid LDH concentration divided by serum LDH concentration greater than 0.6,

(iii) pleural fluid LDH concentration greater than two thirds of the upper limit of normal serum LDH concentration, and

(iv) a pH less than 7.0.

MANAGEMENT

- **Thoracentesis and culture sensitivity–based** antibiotic therapy are appropriate and generally successful for stage I parapneumonic effusions but not stage II effusions or stage III empyema.
- ***Drainage of the pleural space*** by radiologically guided catheters or surgical drainage must be employed for successful management of empyema.
- ***Tube thoracostomy*** using a large-bore chest tube (32–38 Fr) was the initial intervention when empyema was established.
- After chest tube placement, fibrinolytics are administered until the pleural space is cleared radiographically and the patient's clinical condition is improved.
- Streptokinase initially was used. Subsequently, urokinase was used and found to be more efficacious

SURGICAL MANAGEMENT

- About 20–30% require a surgical drainage procedure:
- **I) Thoracoscopy** and debridement should be the next therapeutic maneuver after attempted fibrinolysis. VATS must accomplish two therapeutic goals to be successful: (1) establish a unified pleural space and (2) ensure total reexpansion of the lung parenchyma with obliteration of the pleural cavity
- **II) Decortication** via a thoracotomy should be performed when the third or fibrotic stage of empyema is suggested by CT scan that reveals visceral pleural enhancement without fibrin septation in multiple areas of loculation.

Decortication refers to the process of peeling this restrictive fibrous layer from the pleura, promoting lung reexpansion and improving thoracic excursion

III) Open widow thoracostomy sometimes required

Malignant Pleural Effusion

- Up to 22% of pleural effusions are caused by malignant disease.
- Affected patients with advanced neoplastic diseases experience considerable morbidity as a result of these pleural fluid collections.

ETIOLOGY

- This may be due to primary or secondary tumors of the pleura with seeding of the intrapleural space and lymphatic obstruction.
- Of pleural neoplasms 95% arise from a metastatic source, with lung and breast carcinoma accounting for 75% of all cases.
- Other common causes are lymphoma, gastric cancer, and ovarian cancer.
- Adenocarcinomas of unknown primary constitute a distinct entity in patients in whom the source of the pleural malignancy is never found.
- They are associated with exposure to environmental tobacco smoke.

Diagnostic Procedures

- The diagnosis of pleural effusion is confirmed by thoracentesis. Practically all malignant pleural effusions are exudates. Criteria sets to classify effusions as exudative
- ***Pleural Cytology and Biopsy***
  Cytological evaluations with standard staining (Papanicolaou's smears) generally confirm malignant pleural effusions.
- When standard pleural effusion cytology is nondiagnostic, pleural needle biopsy is indicated

TREATMENT

- The optimal treatment for malignant pleural effusion is controversial.
- Treatment options can be classified into those that rely on drainage alone or those that also obliterate the pleural space.
- Since there is an inflammatory response associated with most pleural effusions, persistent and complete pleural drainage may achieve pleural fusion.

Clinical features *Physical Findings:*

- Patients with malignant pleural effusions are typically symptomatic and complain of dyspnea, cough, or chest pain.
- The discomfort often is independent of respiration and but is aggravated by activity.
- With invasive pleural metastases, the affected nerve roots may radiate pain.
- Physical findings indicating pleural effusion are decreased tactile fremitus and dullness to percussion on examination of the posterior chest

Imaging

- 1) Early evidence of pleural effusion by chest radiograph is costodiaphragmatic sulcus blunting caused by as little as 125 ml of fluid.
- 2)Computed tomography (CT) multiple pleural nodules and nodular pleural thickening are generally limited to effusions of malignant etiology.
- 3)Thoracic ultrasound shows the optimal site for diagnostic thoracentesis of a small pleural effusion. It also helps select ideal entry points for thoracoscopes or pleural biopsy instruments by avoiding adhesions.

Chylothorax

- **Defintion :** Chylothorax is the presence of lymph in the pleural space.
- Chylothorax may be iatrogenic, traumatic, or spontaneous because of congenital or acquired disorders.
- Early recognition is crucial to preventing serious complications of chronic lymphatic loss.
- Initial drainage, combined with measures to reduce chyle flow and maintain nutrition, may resolve the condition.

Chylothorax

- For cases that do not show prompt response, particularly in the postoperative or trauma setting, early surgical ligation of the thoracic duct should be performed with the expectation of prompt resolution.
- Thoracoscopy offers a minimally invasive approach.
- Other alternatives require consideration of the underlying etiology, consideration of the patient's medical condition and prognosis, and an understanding of the limitations of these approaches.

Tumors of the Pleura: Malignant Pleural:

- Primary tumors of the pleura are very rare. There are two main types:

1. **Malignant Pleural Mesothelioma (MPM):** most common primary tumor of the pleura. Highly aggressive tumor with grave prognosis.
2. **Localized Fibrous Tumor of the Pleura:** significantly less aggressive than MM, also referred to as "localized mesothelioma."

- **Epidemiology:** Strong association with asbestos exposure (>80 % o cases), with 20-years latency until the development of MM. 5:1 Male predominance, with peak incidence in the sixth and seventh decade. Other causes: radiation, mineral fibers, simian virus 40.
- **Clinical Presentation**
- Dyspnea
- Pain secondary to chest wall invasion and involvement of somatic nerves of the parietal pleura
- Malignant pleural effusion
- Pleural thickening, subtle pleural masses on radiographic images, Thick, restrictive pleural rind (late finding) with encasement of the lung

Work-up

- **Work-Up**
- CT chest with IV contrast +/− FDG-PET
- MRI as an adjunct to determine the depth of chest wall invasion
- or diaphragmatic involvement
- Thoracocentesis of malignant effusion: cytology and elevated pleural hyaluronic acid have a 50 % diagnostic accuracy.
- Gold standard for diagnosis is thoracoscopic pleural biopsy of about 80 % Diagnostic accuracy
- **Management**
- • Most patients present with locally advanced disease and not amenable to surgical treatment options.
- • Surgery: *Extra pleural Pneumonectomy (EPP):* en-bloc resection of the lung, visceral pleura, parietal pleura, pericardium, hemidiaphragm. Surgery is reserved for patients with localized and selected locally advanced disease. Surgery is provided in the context of multi-modality therapy (neoadjuvant or adjuvant chemotherapy and adjuvant whole thorax radiation therapy), with improvement in overall survival and disease-free survival depending on tumor histopathology and regimen.